

06_Workflow_Model

Workflow Model

HCAT Insight — Rassoul Azam Hospital

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1. Introduction

This document describes the operational workflow of **HCAT Insight** in full detail — how complaints move through the system, who acts at each stage, what decisions are made, and how the workflow branches for different complaint types.

The HCAT Insight workflow is not a simple linear pipeline. It is a multi-branch, role-scoped process that handles ordinary complaints, Red Flag escalations, Never Events, and seasonal organizational reporting through distinct but interconnected paths. Understanding this workflow is essential for understanding both the system's operational value and its technical design.

The workflow was derived directly from the implemented codebase: the database layer (`backend/api/db_layer/`), the router definitions (`backend/api/routers/`), and the role and policy constants (`backend/core/constants/`).

2. Organizational Structure

HCAT Insight models the hospital's organizational hierarchy as a three-tier tree, each tier corresponding to a level of administrative responsibility within the complaint workflow.

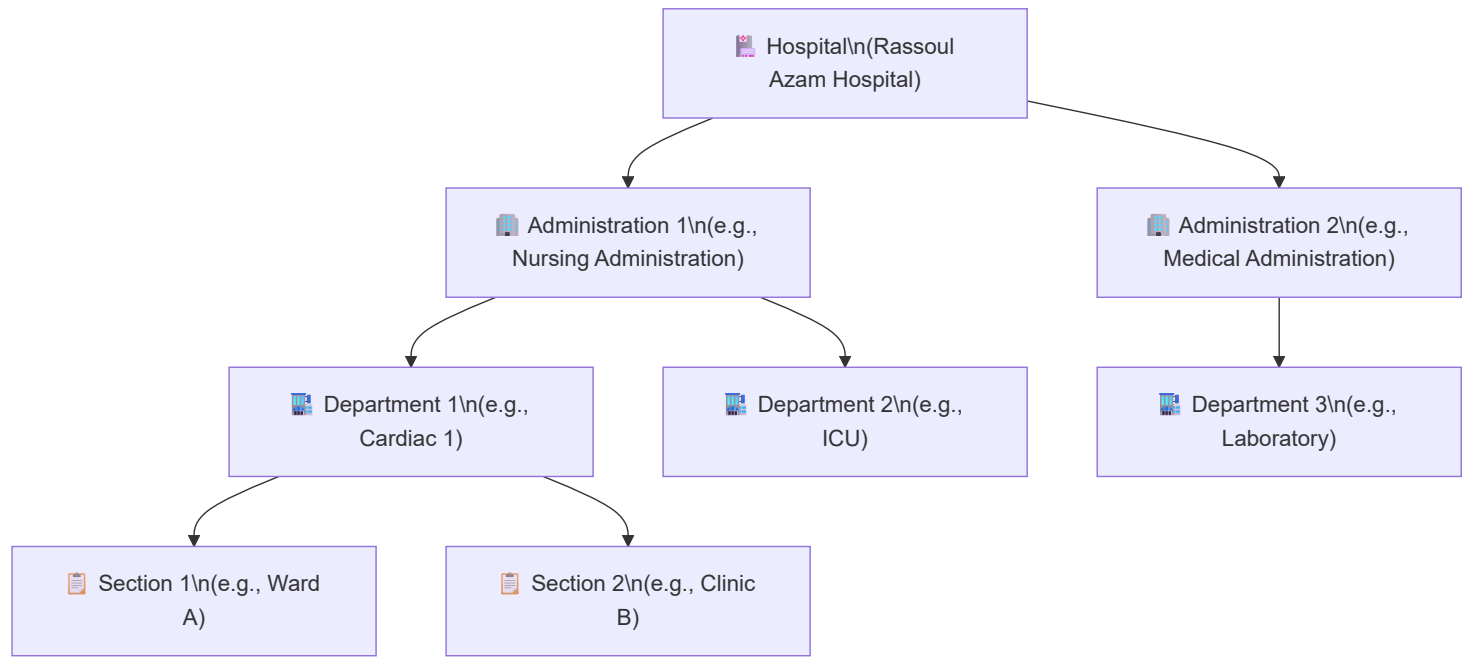


Figure 1: Organizational Hierarchy Model.

Each node in this hierarchy has:

- A designated **administrator** responsible for complaints affecting that unit
- A set of configurable **severity and domain thresholds** that define alert levels
- A **scope of visibility** — administrators at each level can only see complaints relevant to their unit and below

The complaint routing system uses this hierarchy to determine which administrator receives a complaint subcase, which policy applies, and which reports are generated.

3. Role Responsibilities in the Workflow

Each of the seven system roles has defined responsibilities at specific stages of the complaint workflow.

Role	Workflow Stage	Key Responsibilities
WORKER	Intake, Action Execution	Enter complaints, execute assigned action items, basic case data entry
COMPLAINT_SUPERVISOR	Review, Classification, Follow-Up	Review complaints, confirm AI classification, document investigation, follow up with patients

Role	Workflow Stage	Key Responsibilities
SECTION_ADMIN	Subcase Response, RCA	Receive subcases in inbox, submit Root Cause Analysis, provide department explanation
DEPARTMENT_ADMIN	Subcase Oversight, Approval	Oversee subcases within department, approve completed subcases
ADMINISTRATION_ADMIN	Final Approval, Reporting	Final approval of resolved cases, seasonal reporting, cross-department visibility
UNIVERSAL_SECTION	Bridging Role	Handles all section-level approvals during early adoption; sees all subcases without scope filter
SOFTWARE_ADMIN	System Management	User management, system settings, force-close oversight, ML retraining

4. Complaint Intake Workflow

The intake workflow begins when a complaint is received through any of the eight supported channels and ends when the case is formally opened in the system with all required fields populated.

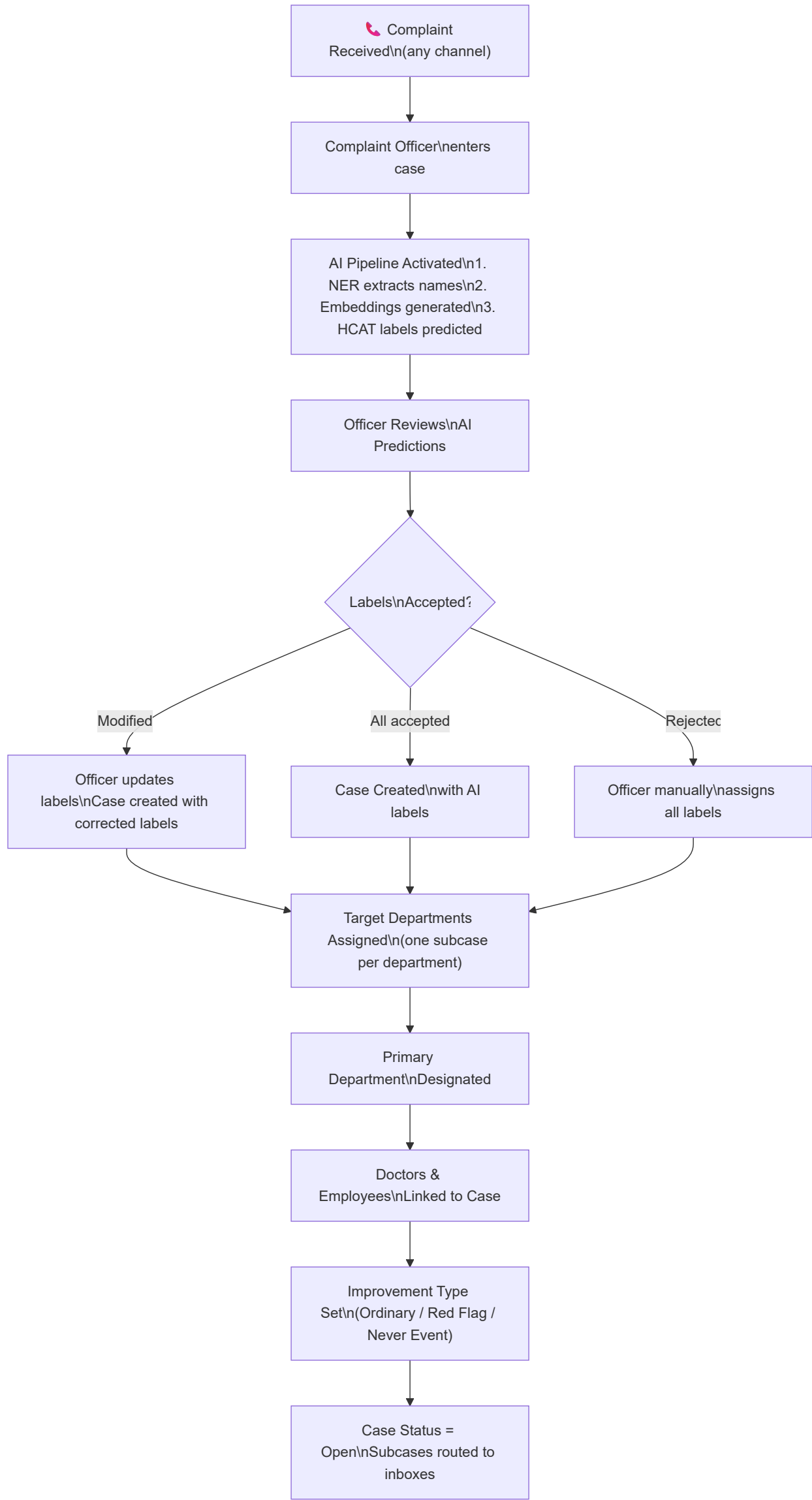


Figure 2: Complaint Intake Workflow.

Key intake decisions:

Decision Point	Options	System Effect
Patient type	Inpatient / Outpatient	Linked to admission record via HIS view

Decision Point	Options	System Effect
Intake channel	8 channels	Recorded as SourceID; affects reporting segmentation
HCAT classification	Accept AI / Modify / Manual	Determines labels applied to case
Target departments	One or multiple	One subcase created per target department
Primary department	Designated from targets	Receives priority in reporting
Improvement type	Ordinary / Red Flag / Never Event	Triggers different downstream workflows
Requires explanation	Yes / No	Activates explanation request in parallel

5. Case and Subcase Lifecycle

5.1 Case States

A Case in HCAT Insight progresses through three lifecycle states:

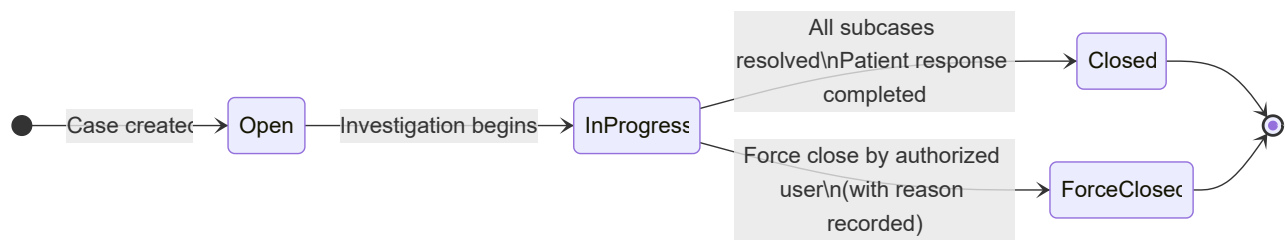


Figure 3: Case Status State Machine.

Status	Description	Who Can Advance
Open	Case created, pending assignment	System (auto)
In Progress	Active investigation or follow-up	Complaint Supervisor, Admin roles
Closed	Fully resolved; all subcases complete	Complaint Supervisor, Admin roles
Force Closed	Administratively closed before natural resolution; reason required	SOFTWARE_ADMIN

5.2 Case Data Structure

Each case record contains:

Identity fields: PatientName, FeedbackReceivedDate, IssuingOrgUnitID, SourceID, isINPatient (inpatient/outpatient flag)

Clinical classification: DomainID, CategoryID, SubCategoryID, ClassificationID, SeverityID, StageID, HarmLevelID, ClinicalRiskTypeID

Operational fields: CaseStatusID, FeedbackIntentTypeID, BuildingID, RequiresExplanation, ExplanationStatusID

Text content: ComplaintText (patient narrative), ImmediateAction (immediate response), TakenAction (detailed actions taken)

Audit fields: CreatedByUserID, ForceClosedAt, ForceClosedByUserID, ForceCloseReason

5.3 Subcase Structure

Each target department assignment creates one **subcase** (APP_IncidentCaseTargetDepartment) linked to the parent case. A case with three target departments generates three subcases, each with its own approval workflow.

Subcase Property	Description
IsPrimary	Exactly one subcase is designated as primary per case
DepartmentID	The target department this subcase is directed to
AssignedByUserID	The user who assigned this department target
AssignedAt	Timestamp of assignment

6. Subcase Approval Chain

Each subcase follows its own approval chain, independent of other subcases in the same case. The approval chain reflects the organizational hierarchy: complaints pass through the Section level before reaching the Administration level.

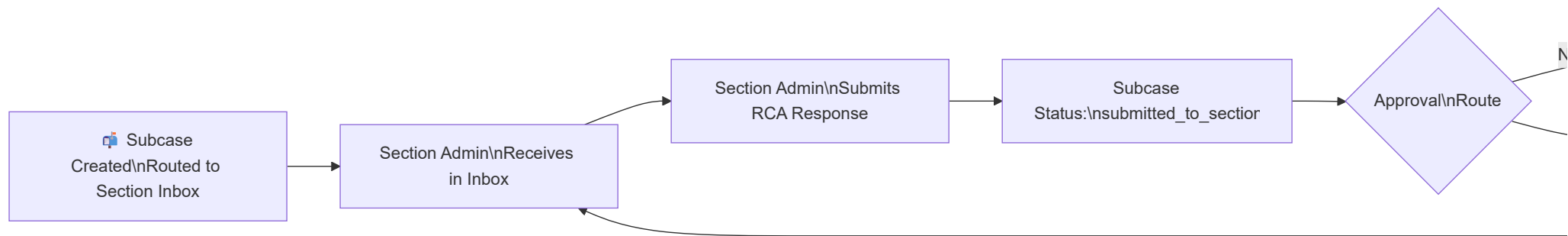


Figure 4: Subcase Approval Chain.

Approval shortcut — UNIVERSAL_SECTION role:

During the early adoption phase, when individual Section Admins may not yet be fully operational in the system, the UNIVERSAL_SECTION role can directly advance subcases from submitted_to_section to admin_approved , bypassing the intermediate Department Admin step. This role sees all subcases across all sections without scope filtering, acting as an operational bridge until the full role structure is in place.

7. Root Cause Analysis Workflow

When a Section Administrator responds to a subcase in their inbox, they are required to complete a **Root Cause Analysis (RCA)** before the subcase can advance to the approval stage. The RCA is stored in APP_IncidentCaseFeedback linked to the specific subcase.

7.1 RCA Categories

The RCA framework structures the cause analysis across four domains:

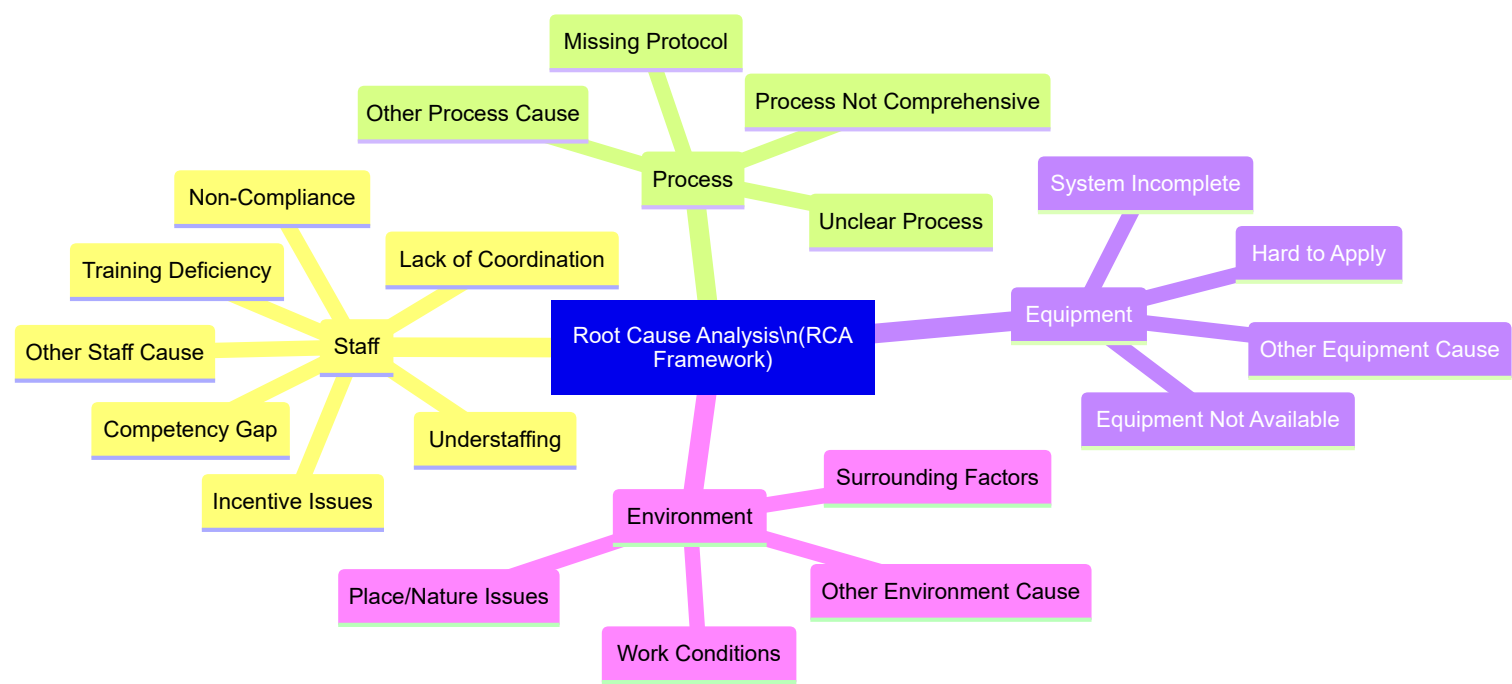


Figure 5: RCA Cause Analysis Framework.

7.2 Preventive Actions

Alongside cause identification, the RCA requires documentation of planned preventive actions:

Preventive Action	Description
Monthly Meetings	Schedule meetings to address the root cause
Training Programs	Implement staff training
Increase Staffing	Request additional personnel
M&M Committee Actions	Submit to Morbidity & Mortality committee
Other	Custom preventive action with free text

7.3 Department Explanation

After completing the RCA, the section may also provide a **Department Explanation** — a formal written response to the complaint, which is tracked with:

- `DepartmentExplanationText` : the full written explanation
- `DepartmentExplanationStatusID` : workflow state (Waiting → Submitted → Reviewed)
- `DepartmentExplanationReceivalDate` : when the explanation was officially received

8. Action Item Workflow

Action items are discrete tasks created in response to a complaint case, a seasonal report, or a seasonal case. They represent concrete corrective or preventive steps that must be executed and tracked.

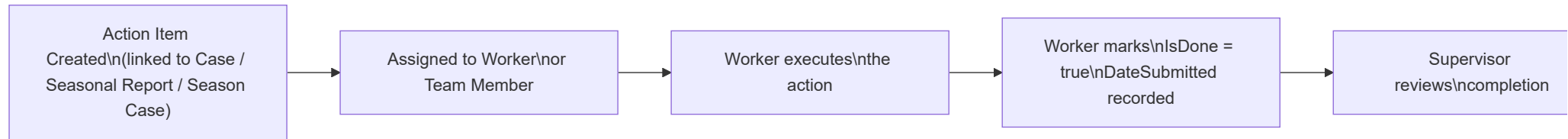


Figure 6: Action Item Workflow.

Action item properties:

- `ActionTitle` : Concise task name
- `ActionDescription` : Detailed task description
- `DueDate` : Required completion date
- `IsDone` : Completion flag
- `DateSubmitted` : Auto-stamped when marked done
- **Parent linkage** (exactly one):
 - `IncidentRequestCaseID` → linked to a complaint case
 - `SeasonalReportID` → linked to a seasonal report
 - `SeasonCaseID` → linked to a specific department's seasonal case

Action items can accumulate per case over its lifecycle and are visible in the case detail view, providing an audit trail of all corrective actions taken.

9. Follow-Up and Patient Communication Workflow

HCAT Insight tracks follow-up contacts with the patient or guardian throughout the complaint lifecycle. The follow-up module (`follow_up_db.py` , `follow_up_router.py`) records:

- Date and method of contact
- Summary of communication
- Patient/guardian response
- Status of the follow-up (pending, completed, unsuccessful contact)

Follow-up records are attached to the parent case and visible to the Complaint Supervisor and above. Multiple follow-up contacts may be recorded for a single case, providing a complete communication timeline.

Follow-up requirement: Cases with the `RequiresExplanation` flag set must include a formal follow-up record confirming that the patient/guardian received and acknowledged the hospital's formal explanation before the case can be closed.

10. Explanation and Reporting Workflow

HCAT Insight manages three distinct explanation tracks, reflecting the three complaint priority types:

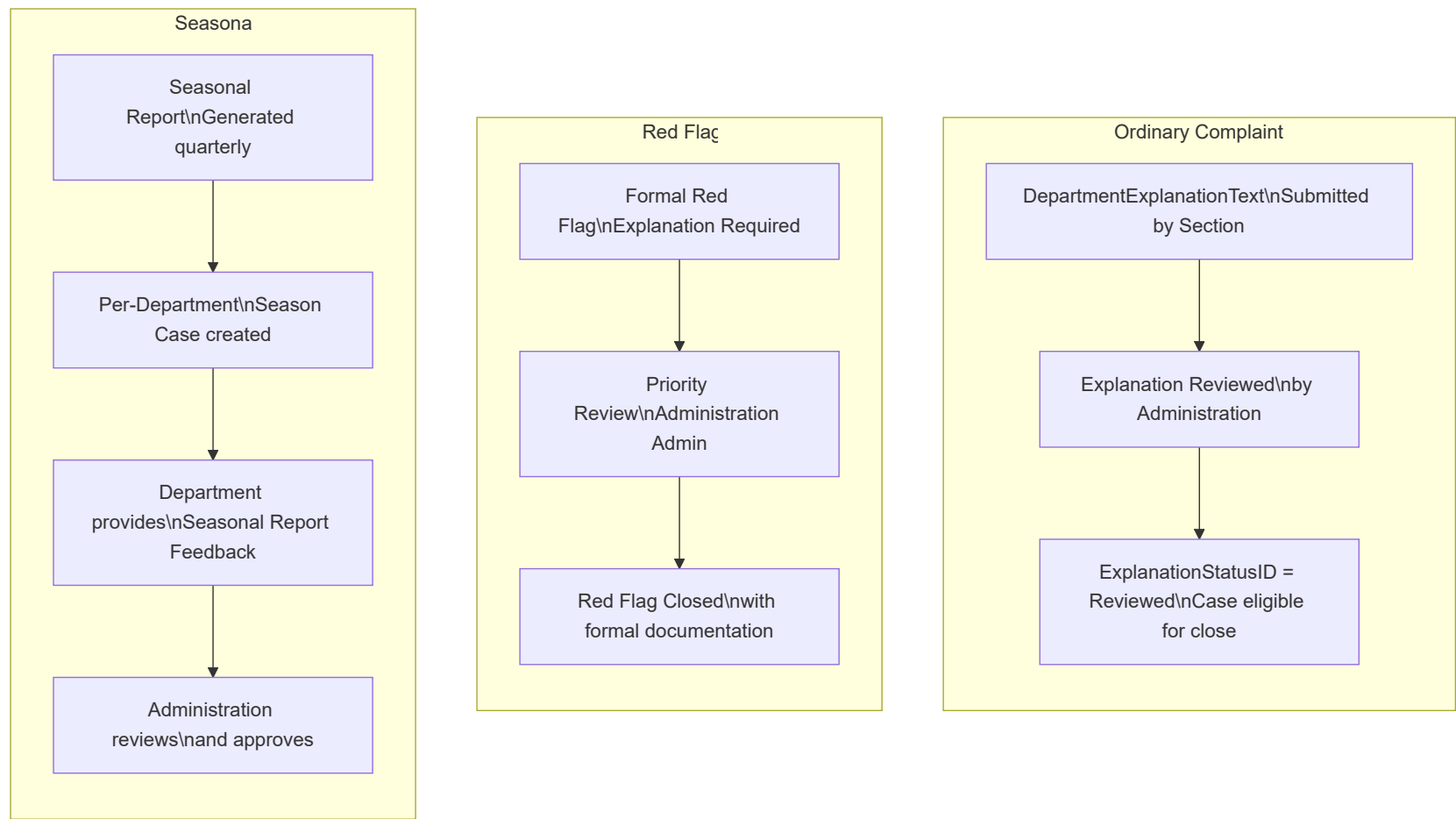


Figure 7: Explanation Workflow Branches by Complaint Type.

The `explanation_ordinary_db.py`, `explanation_red_flag_db.py`, and `explanation_seasonal_db.py` database layers implement these three separate explanation tracks, each with their own status progression and approval requirements.

11. Red Flag and Never Event Workflow

Red Flag and Never Event cases follow accelerated and enhanced workflows compared to ordinary complaints.

11.1 Red Flag Workflow

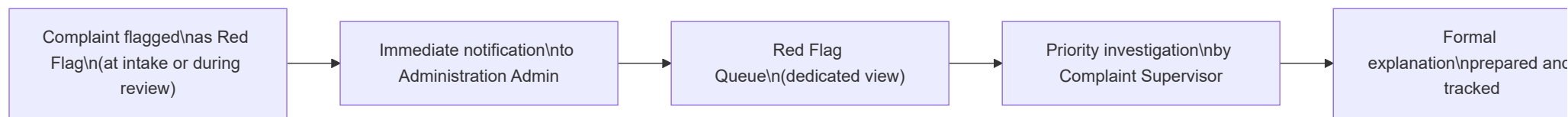


Figure 8: Red Flag Workflow.

11.2 Never Event Workflow

Never Events represent the highest severity category — events that should never occur in a properly functioning healthcare environment.

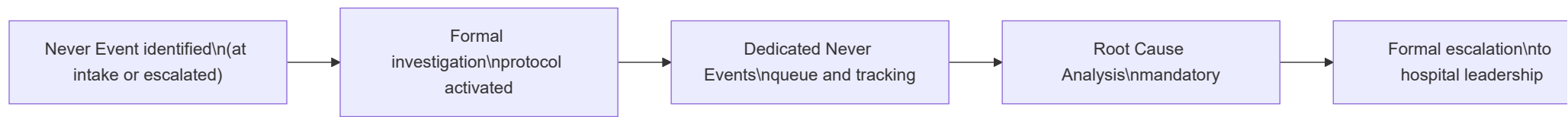


Figure 9: Never Event Workflow.

Both Red Flag and Never Event cases are accessible through dedicated router endpoints (`red_flags_router.py` , `never_events_router.py`) and appear in dedicated filtered views in the frontend, ensuring they are never buried within the general complaint queue.

12. Seasonal Reporting Workflow

In addition to the per-complaint workflow, HCAT Insight operates a parallel **seasonal reporting cycle** that generates quarterly organizational reports for management.

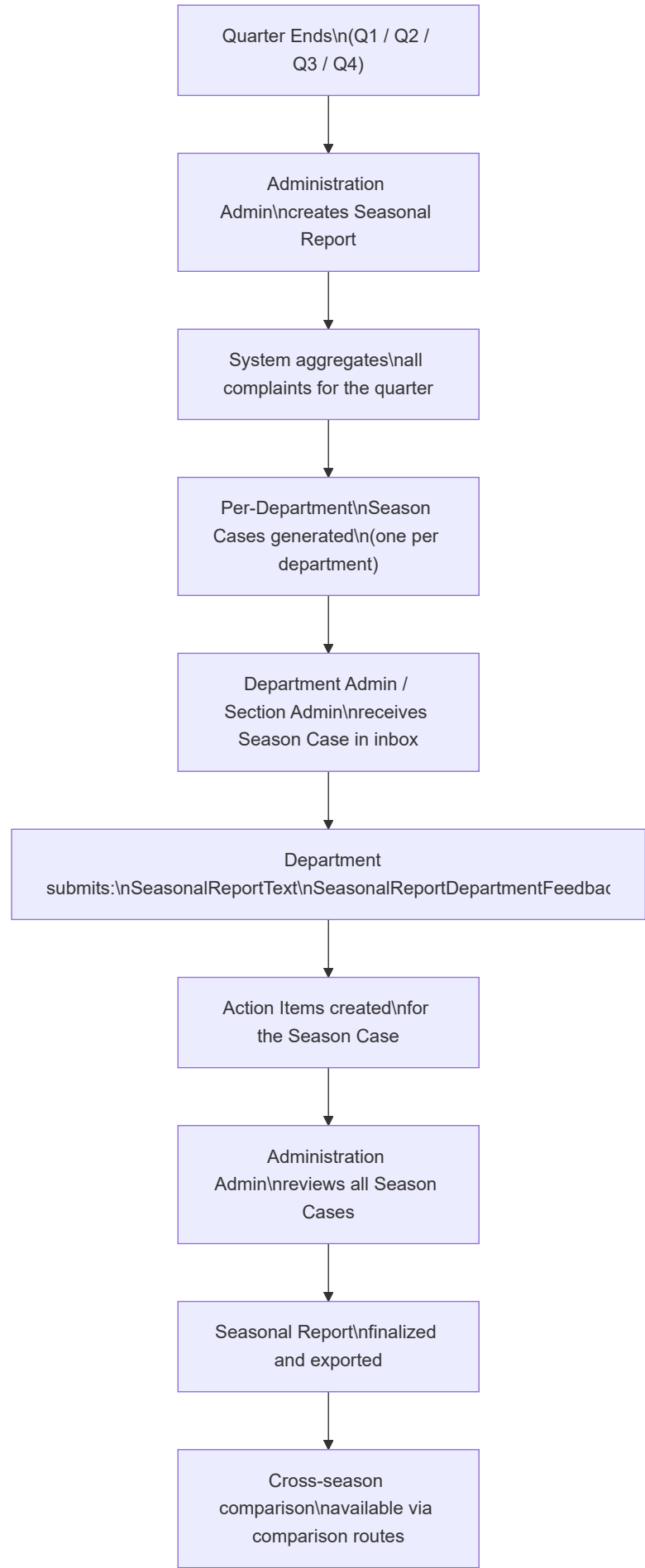


Figure 10: Seasonal Reporting Workflow.

Seasonal action items: Seasonal cases have their own action items (`SeasonCaseID` linkage), separate from per-complaint action items. This allows corrective actions arising from quarterly analysis to be tracked independently from individual case resolution actions.

Export functionality: Finalized seasonal reports can be exported via `seasonal_export_router.py` , enabling offline distribution to stakeholders who do not have system access.

13. Organizational Policy and Thresholds

Each organizational unit (Administration, Department, or Section) has a configurable **policy** stored in `APP_OrgUnitPolicy` that defines alert thresholds. These policies drive automatic flagging and reporting logic.

Policy Parameter	Description
<code>LowSeverityLimit</code>	Maximum acceptable number of Low severity complaints per period
<code>MediumSeverityLimit</code>	Maximum acceptable number of Medium severity complaints per period
<code>HighSeverityLimit</code>	Maximum acceptable number of High severity complaints per period
<code>ClinicalDomainLimit</code>	Threshold for complaints in the CLINICAL domain
<code>ManagementDomainLimit</code>	Threshold for complaints in the MANAGEMENT domain
<code>RelationalDomainLimit</code>	Threshold for complaints in the RELATIONAL domain
<code>EnableLowSeverityRepetitionRule</code>	Flag repeated Low severity complaints as patterns
<code>EnableMediumSeverityRepetitionRule</code>	Flag repeated Medium severity complaints
<code>EnableHighSeverityPercentageRule</code>	Alert when High severity exceeds a percentage of total
<code>EnableHighSeverityPercentageByDomainRule</code>	Alert when High severity in a specific domain exceeds threshold

These thresholds enable HCAT Insight to move beyond simple complaint counting toward **pattern-aware quality monitoring** — alerting administrators when complaint patterns indicate systemic issues rather than isolated incidents.

14. Visibility and Scope Rules

Each role has a defined scope of visibility that controls which cases, subcases, and reports it can access.

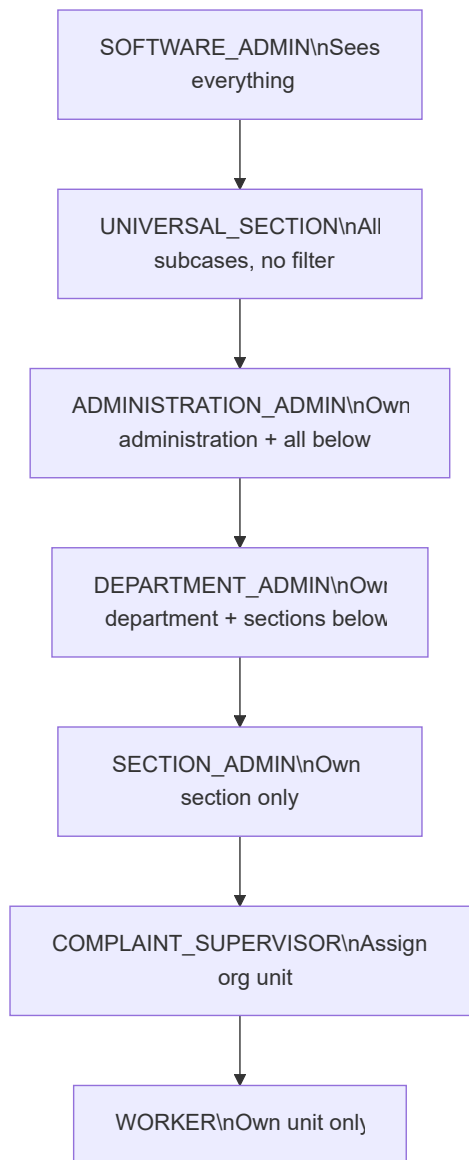


Figure 11: Scope Hierarchy for Data Visibility.

Scope enforcement: The backend enforces scoping at the database query level through dynamic WHERE clauses that filter by `IssuingOrgUnitID` against the authenticated user's organizational unit assignments. A Section Admin cannot access cases from a different section, even if they share the same department.

Scope override: The UNIVERSAL_SECTION role bypasses scope filtering entirely ("no scope filter" as noted in the role code comment), allowing it to see all subcases across all sections. This is intentional for the early adoption period.

15. Workflow Limitations and Operational Challenges

Challenge	Description	Current Mitigation
Multi-department complaint complexity	A single complaint targeting five departments generates five subcases, each requiring separate RCA and approval	System supports this natively; dashboard aggregates status
Section Admin non-participation	If a Section Admin does not log in or respond, their subcase stalls the case	UNIVERSAL_SECTION role bridges the gap during early adoption
Force close requirement	Some cases must be closed administratively when patients cannot be reached	Force close mechanism with mandatory reason and audit trail
Explanation tracking fragmentation	Three separate explanation tracks (ordinary, red flag, seasonal) require different database tables	Separate router and service layers handle each track
Seasonal workflow length	Quarterly reporting requires coordination across many departments	Season cases provide per-department tracking within the same seasonal report
Workflow state visibility	Users at lower role levels cannot see the full case status across all subcases	Dashboard aggregates provide summary visibility

16. Conclusion

The HCAT Insight workflow model is a multi-path, role-scoped complaint management process that operates at two parallel levels:

- Per-complaint workflow:** Individual complaints enter through intake, are AI-classified, split into department-specific subcases, processed through a Root Cause Analysis and approval chain, and closed with documented resolution and patient follow-up.
- Seasonal reporting workflow:** Quarterly, the system aggregates all complaints into organizational reports, generates department-level seasonal cases, and tracks corrective actions and management responses at the organizational level.

The workflow reflects the operational reality of a specialized hospital: complaints frequently involve multiple departments, require structured cause analysis, generate concrete action items, and must be documented in a way that supports both patient communication and organizational learning.

The key design principles are:

- Role-scoped access** ensures each participant sees only what is relevant to their responsibilities
- Subcase independence** allows multi-department complaints to be processed in parallel
- RCA requirement** transforms complaint response from reactive to analytical
- Policy thresholds** move the system from incident tracking to pattern detection
- Dual workflow tracks** (per-complaint and seasonal) ensure both immediate resolution and long-term quality improvement are captured